

Complex Numbers: Intro & Basics

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Outline

- 1 Introduction
- 2 Where Do We Use Complex Numbers In ECE
- 3 Rectangular Form
- 4 Polar Form
- 5 Complex Roots
- 6 Practice Problems

Quick Reminders: Geometry & Complex Numbers

Key Formulas to Remember

- Hypotenuse :

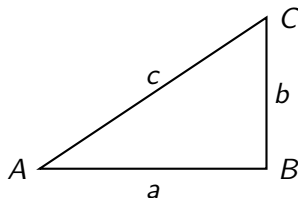
$$c = \sqrt{a^2 + b^2}$$

- Euler's Formula:

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

- Difference of Squares:

$$a^2 - b^2 = (a + b)(a - b)$$



Why Do We Need Complex Numbers?

Real Roots:

$$x^2 - 4 = 0$$

$$x = +2, -2$$

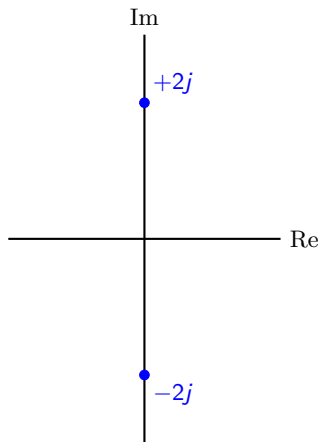


Real roots lie on the number line.

Imaginary Roots:

$$x^2 + 4 = 0$$

$$x = -2j, +2j$$



Definition

A complex number is defined as:

$$z = a + jb, \quad a, b \in \mathbb{R}, \quad j^2 = -1.$$

- Real part: $a = \Re\{z\}$
- Imaginary part: $b = \Im\{z\}$

Argand Diagram

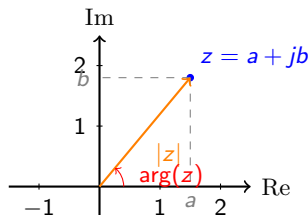
Visualizing Complex Numbers

A complex number $z = a + jb$ corresponds to the point (a, b) in the plane:

- a : Real part (horizontal axis)
- b : Imaginary part (vertical axis)

Key Properties:

- Magnitude: $|z| = \sqrt{a^2 + b^2}$
- Angle: $\arg(z) = \text{atan2}(b, a) = \tan^{-1}\left(\frac{b}{a}\right)$ (quadrant adjustment may be needed)



Complex Numbers in Electrical Engineering

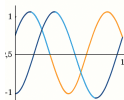
Why do ECE students need complex numbers?

Key Applications:

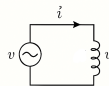
- AC circuit analysis
- Signal processing (Fourier transforms)
- Electromagnetics

Applications of complex numbers

Signal Processing



Electric Circuits



Electromagnetic



Core Idea

Complex numbers turn differential equations into algebra!

Complex Numbers Simplify Calculus

From Differential Equations to Algebra

- Complex exponentials like $e^{j\omega t}$ make derivatives simple:

$$\frac{d}{dt}e^{j\omega t} = j\omega e^{j\omega t}$$

- This turns differential equations into algebraic equations in $j\omega$
- Very useful in circuits, control systems, and signal processing

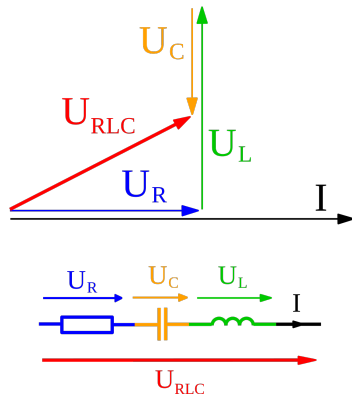
Why It Matters

Using $e^{j\omega t}$, we avoid solving complex calculus—just use algebra to analyze system behavior!

AC Circuit Application

Understanding Phasors in Circuits

- **Phasors** are rotating arrows used to represent voltage and current in AC circuits.
- Instead of showing the wavy shape of the signal, we use a simple rotating line (phasor) that makes calculations easier.
- The angle shows the timing (phase), and the length shows the strength (magnitude) of the signal.



Engineers' Secret Tool

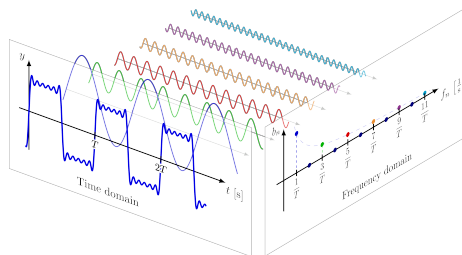
Phasors turn wavy AC signals into simple rotating arrows we can analyze visually

Voltage (blue) and current (red) phasors in an RLC circuit

Signal Processing

Fourier Transform: Decomposing Signals

- Any signal = Sum of **complex exponentials**
- Example:**
Audio signal $\rightarrow$ Frequency spectrum



Time domain (left) vs. frequency domain (right)

Why Complex?

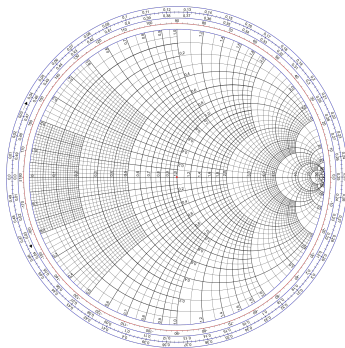
$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

Captures both amplitude and phase!

Impedance Matching

Designing RF Systems with Smith Charts

- Impedance: $Z = R + jX$
- Match Z_{antenna} to $Z_{\text{transmitter}}$
- **Example:**
Match 50Ω to $(75 + j25)\Omega$



Complex impedance visualized on a Smith chart

Key Benefit

Avoids signal reflections.

Complex numbers combine resistance and reactance, making impedance easy to plot and match on the Smith chart.

Addition & Subtraction

If $z_1 = a + jb$ and $z_2 = c + jd$, then:

$$z_1 + z_2 = (a + c) + j(b + d), \quad z_1 - z_2 = (a - c) + j(b - d).$$

Example:

$$(4 + 3j) - (2 - 6j) = 2 + 9j.$$

Multiplication

General case.

$$\begin{aligned}z_1 z_2 &= (a + jb)(c + jd) \\&= ac + a(jd) + (jb)c + (jb)(jd) \\&= ac + jad + jbc + j^2 bd \\&= (ac - bd) + j(ad + bc).\end{aligned}$$

Example:

$$\begin{aligned}(1 + 2j)(4 + 6j) &= 1 \cdot 4 + 1 \cdot 6j + 2j \cdot 4 + 2j \cdot 6j \\&= 4 + 6j + 8j + 12j^2 \\&= 4 + 14j + 12(-1) \\&= (4 - 12) + 14j \\&= -8 + 14j.\end{aligned}$$

Complex Conjugate

For $z = a + jb$, its conjugate is:

$$\bar{z} = a - jb.$$

- $(z_1 + z_2)^* = z_1^* + z_2^*$
- $(z_1 z_2)^* = z_1^* z_2^*$
- $z z^* = |z|^2$
- $(a + jb) * (a - jb) = a^2 - (jb)^2 = a^2 + b^2$

Example: $(-1 + 2j)^* = -1 - 2j$

Division

To divide z_1/z_2 multiply by the conjugate of z_2 :

$$\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} = \frac{(a + jb)(c - jd)}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + j \frac{bc - ad}{c^2 + d^2}.$$

Example:

$$\frac{2 + 4j}{6 - 7j} = \frac{-16 + 38j}{85} = -\frac{16}{85} + j \frac{38}{85}.$$

Division: Step-by-Step

General case.

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{a + jb}{c + jd} \times \frac{c - jd}{c - jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} \\&= \frac{ac - ajd + jbc - jbjd}{c^2 - (jd)^2} \\&= \frac{ac - jad + jbc - j^2 bd}{c^2 + d^2} \\&= \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2} \\&= \frac{ac + bd}{c^2 + d^2} + j \frac{bc - ad}{c^2 + d^2}.\end{aligned}$$

Division: Example

$$\begin{aligned}\frac{2+4j}{6-7j} &= \frac{(2+4j)\overline{(6-7j)}}{(6-7j)\overline{(6-7j)}} = \frac{(2+4j)(6+7j)}{6^2 + (-7)^2} \\&= \frac{2 \cdot 6 + 2 \cdot 7j + 4j \cdot 6 + 4j \cdot 7j}{36 + 49} = \frac{12 + 14j + 24j + 28j^2}{85} \\&= \frac{12 + (14j + 24j) + 28(-1)}{85} = \frac{12 + 38j - 28}{85} = \frac{-16 + 38j}{85} \\&= -\frac{16}{85} + j \frac{38}{85}.\end{aligned}$$

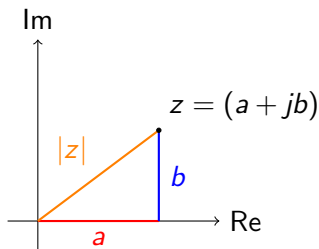
Key step: use the *conjugate* of the denominator, $\overline{6-7j} = 6+7j$, to clear the imaginary part below.

Magnitude

For $z = a + jb$, the magnitude (or modulus) is:

$$|z| = \sqrt{a^2 + b^2}.$$

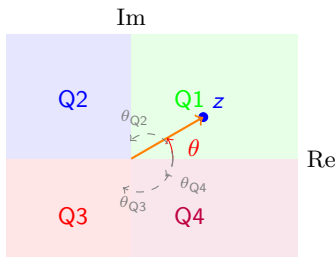
Geometrically, this is the hypotenuse of a right triangle with legs a and b :



Argument (Angle) of a Complex Number

For $z = a + jb$, the argument $\theta = \arg(z)$ is the angle from the positive real axis:

$$\theta = \text{atan2}(b, a)$$



Quadrant Rules:

- Q1 ($a > 0, b > 0$): $\tan^{-1}(b/a)$
- Q2 ($a < 0, b > 0$): $\pi - \tan^{-1}(|b/a|)$
- Q3 ($a < 0, b < 0$): $\pi + \tan^{-1}(b/a)$
- Q4 ($a > 0, b < 0$): $2\pi - \tan^{-1}(|b/a|)$

Visual Key:

- Q1: Green background
- Q2: Blue background
- Q3: Red background
- Q4: Purple background

Key Point

$\text{atan2}(b, a)$ automatically handles quadrants, unlike $\tan^{-1}(b/a)$.

Polar Form of Complex Numbers

Three Equivalent Forms:

① Trigonometric:

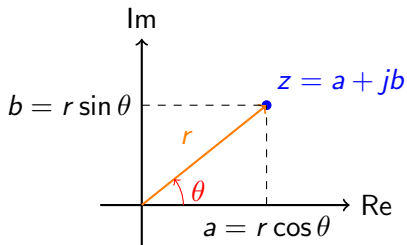
$$z = r(\cos \theta + j \sin \theta)$$

② Polar Notation:

$$z = r \angle \theta$$

③ Exponential (Euler):

$$z = re^{j\theta}$$



Key Components

- $r = |z| = \sqrt{a^2 + b^2}$ (magnitude)
- $\theta = \arg(z)$ (angle in radians)

Multiplication & Division in Polar Form

Given $z_1 = r_1 \angle \theta_1$ and $z_2 = r_2 \angle \theta_2$:

Multiplication Rule

$$z_1 z_2 = (r_1 r_2) \angle (\theta_1 + \theta_2)$$

Example:

$$2 \angle 30^\circ \times 3 \angle 60^\circ = 6 \angle 90^\circ$$

Division Rule

$$\frac{z_1}{z_2} = \left(\frac{r_1}{r_2} \right) \angle (\theta_1 - \theta_2)$$

Example:

$$\frac{8 \angle 120^\circ}{4 \angle 30^\circ} = 2 \angle 90^\circ$$

Note: Angles may need adjustment to stay in $(-\pi, \pi]$ or $[0, 2\pi)$

Addition: Which Form Should We Use?

Question: Add the following two complex numbers:

$$z_1 = 3 + 4j, \quad z_2 = 1 - 2j$$

Rectangular Form

$$z_1 + z_2 = (3 + 4j) + (1 - 2j)$$

Polar Form

- Convert to polar:

$$z_1 = 5\angle 53.13^\circ, \quad z_2 = \sqrt{5}\angle -63.43^\circ$$

Addition: Which Form Should We Use?

Question: Add the following two complex numbers:

$$z_1 = 3 + 4j, \quad z_2 = 1 - 2j$$

Rectangular Form

$$\begin{aligned} z_1 + z_2 &= (3 + 4j) + (1 - 2j) \\ &= (3 + 1) + j(4 - 2) \\ &= 4 + 2j \end{aligned}$$

Polar Form

- Convert to polar:

$$z_1 = 5\angle 53.13^\circ, \quad z_2 = \sqrt{5}\angle -63.43^\circ$$

- Addition in polar is difficult:

Must convert to rectangular first.

Tip

For addition, rectangular form is more convenient and direct.

Multiplication: Which Form Should We Use?

Question: Multiply the following two complex numbers:

$$z_1 = 1 + j, \quad z_2 = \sqrt{3} + j$$

Rectangular Form

$$z_1 z_2 = (1 + j)(\sqrt{3} + j)$$

Polar Form

$$z_1 = \sqrt{2} \angle 45^\circ, \quad z_2 = 2 \angle 30^\circ$$

Multiplication: Which Form Should We Use?

Question: Multiply the following two complex numbers:

$$z_1 = 1 + j, \quad z_2 = \sqrt{3} + j$$

Rectangular Form

- Multiply directly:

$$\begin{aligned} z_1 z_2 &= (1 + j)(\sqrt{3} + j) \\ &= 1 \cdot \sqrt{3} + 1 \cdot j + j \cdot \sqrt{3} + j^2 \\ &= \sqrt{3} + j + j\sqrt{3} - 1 \\ &= (\sqrt{3} - 1) + j(1 + \sqrt{3}) \end{aligned}$$

Polar Form

- Convert first:

$$z_1 = \sqrt{2} \angle 45^\circ, \quad z_2 = 2 \angle 30^\circ$$

- Multiply:

$$z_1 z_2 = 2\sqrt{2} \angle 75^\circ$$

Tip

Polar form makes multiplication faster, especially with angle and magnitude interpretation.

Finding n th Roots of Complex Numbers

General Formula: For $z = re^{j\theta}$, the n th roots are:

$$z_k = r^{1/n} \exp \left(j \frac{\theta + 2\pi k}{n} \right), \quad k = 0, 1, \dots, n-1$$

Key Properties:

- n distinct roots equally spaced on a circle
- Radius of circle: $r^{1/n}$
- Angle between roots: $2\pi/n$

Example 1: 4th Roots of -16

Step 1: Convert to polar form

$$-16 = 16e^{j\pi}$$

Step 2: Apply root formula

$$z_k = 16^{1/4} \exp\left(j\frac{\pi + 2\pi k}{4}\right) = 2e^{j(\pi/4 + k\pi/2)}$$

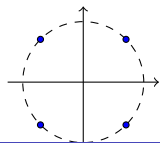
Step 3: Find all roots (k=0 to 3)

$$z_0 = 2e^{j\pi/4} = 2\left(\cos\frac{\pi}{4} + j\sin\frac{\pi}{4}\right)$$

$$z_1 = 2e^{j3\pi/4} = \dots$$

$$z_2 = 2e^{j5\pi/4} = \dots$$

$$z_3 = 2e^{j7\pi/4} = \dots$$



Example 2: 5th Roots of Unity

Step 1: Start with 1 in polar form

$$1 = 1e^{j0}$$

Step 2: Apply root formula

$$z_k = \exp\left(j\frac{2\pi k}{5}\right), \quad k = 0, \dots, 4$$

Step 3: Explicit roots

$$z_0 = e^{j0} = 1$$

$$z_1 = e^{j2\pi/5} \approx 0.309 + j0.951$$

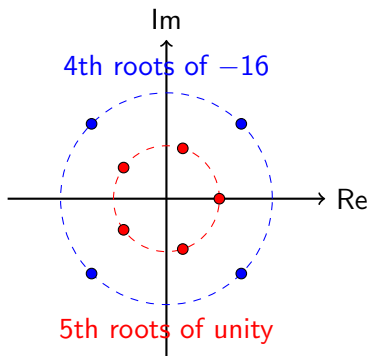
$$z_2 = e^{j4\pi/5} \approx -0.809 + j0.588$$

$$z_3 = e^{j6\pi/5} \approx -0.809 - j0.588$$

$$z_4 = e^{j8\pi/5} \approx 0.309 - j0.951$$



Visualizing Roots on Complex Plane



- Roots lie on a circle in complex plane
- Equally spaced at angles of $2\pi/n$
- First root starts at θ/n (or 0 for unity)

Kahoote Time

<https://kahoot.it/> With Supersizes for 2 winners.

Practice Exercises

Try solving these problems:

- 1 Compute $(2 - 3j)(4 + 5j) + 1/j$
- 2 Express $1 - \sqrt{3}j$ in polar form and find its cube
- 3 Find all 4th roots of -16
- 4 Simplify $\frac{(3+4j)(1-2j)}{2-j}$ in phasor form

After attempting, see next slide for detailed solutions

Practice Solutions

❶ **Compute** $(2 - 3j)(4 + 5j) + 1/j$

$$\begin{aligned} &= (2 \cdot 4 + 2 \cdot 5j - 3j \cdot 4 - 3j \cdot 5j) + (-j) \\ &= (8 + 10j - 12j - 15j^2) - j \\ &= (8 - 2j + 15) - j \quad (\text{since } j^2 = -1) \\ &= 23 - 3j \end{aligned}$$

❷ **Polar form of** $1 - \sqrt{3}j$ **and its cube**

$$\begin{aligned} r &= \sqrt{1^2 + (-\sqrt{3})^2} = 2 \\ \theta &= \tan^{-1}(-\sqrt{3}/1) = -\pi/3 \\ \text{Polar form} &= 2e^{-j\pi/3} = 2\angle -\frac{\pi}{3} = 2\angle -60^\circ \\ \text{Cube} &= (2)^3 e^{j3(-\pi/3)} = 8e^{-j\pi} = -8 \end{aligned}$$

Practice Solutions (Continued)

3 4th roots of -16

$$-16 = 16e^{j\pi}$$

$$\begin{aligned} z_k &= 16^{1/4} e^{j(\pi+2\pi k)/4} \quad k = 0, 1, 2, 3 \\ &= 2e^{j(\pi/4+k\pi/2)} \end{aligned}$$

$$\text{Roots: } 2e^{j\pi/4}, 2e^{j3\pi/4}, 2e^{j5\pi/4}, 2e^{j7\pi/4}$$

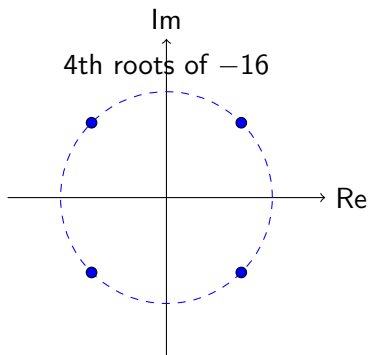
4 Simplify $\frac{(3+4j)(1-2j)}{2-j}$

$$\begin{aligned} \text{Numerator} &= (3 \cdot 1 + 3 \cdot (-2j) + 4j \cdot 1 + 4j \cdot (-2j)) \\ &= (3 - 6j + 4j - 8j^2) = (11 - 2j) \end{aligned}$$

$$\text{Denominator} = 2 - j$$

$$\begin{aligned} \text{Multiply by conjugate} &= \frac{(11 - 2j)(2 + j)}{(2 - j)(2 + j)} \\ &= \frac{22 + 11j - 4j - 2j^2}{4 + 1} = \frac{24 + 7j}{5} \\ &= 4.8 + 1.4j \end{aligned}$$

Visualization of Roots



- Roots are equally spaced on a circle of radius 2
- Angle between roots is 90° ($\pi/2$ radians)
- First root at 45° ($\pi/4$ radians)

Thank You!

Questions?

Slides and notes at: <https://sites.google.com/ucsd.edu/complex-numbers-tutorial/complex-numbers-tutorial?authuser=1>